Does implied volatility efficiently predict future realized volatility?

And how does the answer differ between Bitcoin and the S&P 500?

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The research question

Does implied volatility predict future realized volatility?

Same answer for Bitcoin and the S&P 500?

- Implied volatility is the market's forecast of future risk.
- In equities, the relationship has been studied for thirty years.
- In crypto, it has barely been documented.

What we know vs. what we don't

Equity markets

- Implied volatility is informative
  Christensen & Prabhala (1998)
- Slope below one, positive premium
  Bollerslev, Tauchen & Zhou (2009)
- Predictive power weakens in crises
  Whaley (2009), Giot (2005)

Crypto markets

- DVOL exists only since March 2021
- No published predictive evidence
- No regime-conditional analysis of BTC option markets

Our contribution: apply the equity benchmark to crypto, in a unified framework, with regime splits.

Five hypotheses

- **H1.** Implied volatility overestimates future realized volatility on average: positive variance risk premium.
- **H2.** Implied volatility is informative but not a perfect predictor.
- **H3.** The variance risk premium is larger in crypto than in equities.
- **H4.** The predictive relationship weakens in stress periods.
- **H5.** Equity markets are close to efficient in normal periods; crypto is not.

Data

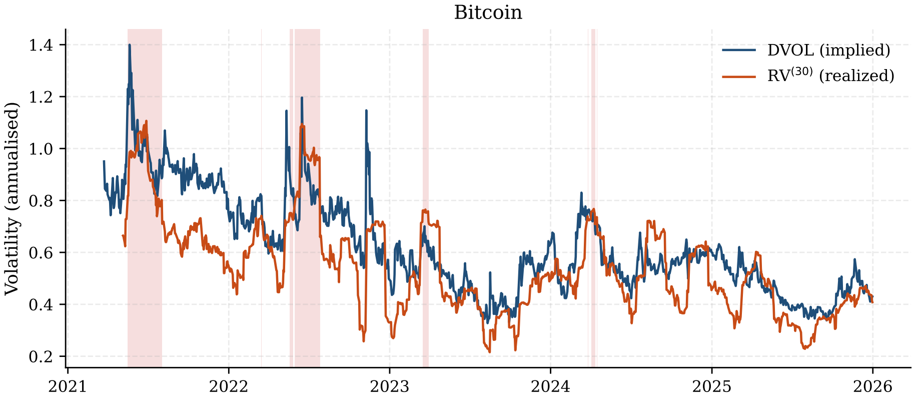
Series	Provider	Identifier
Bitcoin spot	Binance public API	`BTCUSDT` daily
Bitcoin implied vol.	Deribit public API	`BTC_DVOL`
S&P 500 close	Yahoo Finance (`yfinance`)	`GSPC`
VIX close	Yahoo Finance (`yfinance`)	`VIX`

- **Period:** 24 March 2021 $\rightarrow$ 31 December 2025.
- **After NYSE-calendar inner join:** 1 201 daily observations.
- All series come from public APIs (no proprietary terminal data).

Method, in one slide

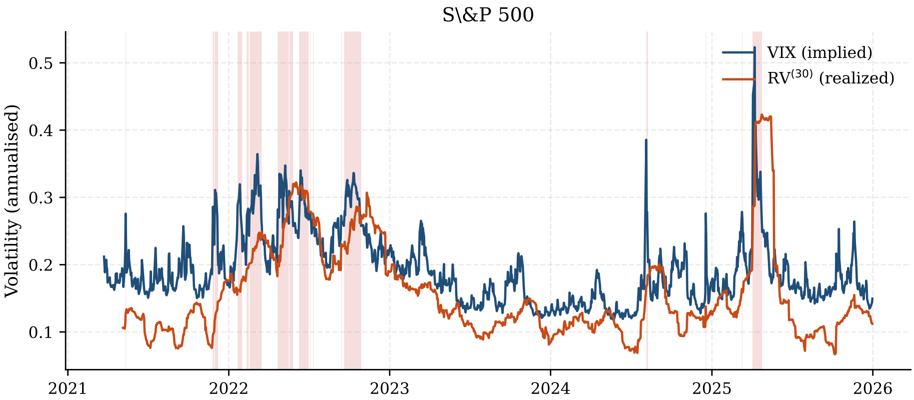
- **Realized volatility:** annualised standard deviation of 30-day log returns.
- **Predictive regression:** future realized volatility regressed on current implied volatility.
  Slope $= 1$ *means efficient.*
- **Regimes:** *stress* $=$ top 10 % of VIX (equities) or BTC realized volatility (crypto), set ex ante.
- **Inference:** Newey–West HAC standard errors to correct for rolling-window overlap.

A first look: Bitcoin



- DVOL and 30-day realized volatility move together, with a positive average wedge.
- Red shading: stress days (BTC 30-day RV above its 90th percentile, set ex ante).

A first look: S&P 500



- Volatility runs in a much narrower band than Bitcoin, but spikes sharply in crises.
- Red shading: stress days (VIX above its 90th percentile, set ex ante).

Implied volatility is informative, but biased

	Bitcoin	**S&P 500**
Slope $\hat{\beta}$	**0.55*****	**0.79*****
HAC s.e.	(0.09)	(0.10)
R^2	0.33	0.36
$H_0 : \beta = 1$	rejected ($p < 0.001$)	rejected ($p = 0.03$)
N	1 171	1 171

- Both slopes are highly significant: implied volatility carries real information.
- Both are significantly below one: positive variance risk premium in both markets.

*** *significant at 1 %. Standard errors: Newey–West HAC, 30 lags.*

The variance risk premium is larger in crypto

Bitcoin

+0.081

avg. variance risk premium

vs.

S&P 500

+0.035

avg. variance risk premium

Crypto investors require more than twice the compensation per unit of volatility risk.

Confirms H3.

Headline result

In calm markets, the VIX looks almost efficient.

$$\hat{\beta} = 0.97$$

S&P 500, normal regime, $N = 955$

$$p = 0.89 \text{ for } H_0 : \beta = 1$$

We cannot reject that the VIX is a fully efficient predictor of S&P 500 realized volatility in normal periods.

The premium reacts to stress in opposite directions

	Normal	Stress	Direction
Bitcoin	$+0.087$	$+0.055$	*compresses*
S&P 500	$+0.032$	$+0.047$	*widens*

- **Equity stress** $\Rightarrow$ premium *widens*. Implied volatility surges with hedging demand.
- **Crypto stress** $\Rightarrow$ premium *compresses*. Realized volatility catches up faster than DVOL.

An asymmetry only visible when both markets are studied together.

Why? Three economic mechanisms

1. **Hedging demand.** Equity investors buy out-of-the-money puts in stress, which pushes the VIX above expected realized volatility.
2. **24/7 trading.** Bitcoin realized volatility adjusts to news within hours, not days, which compresses the wedge faster.
3. **Market maturity.** Equity option markets are deep and dominated by institutions; crypto options are concentrated on a few venues and a more retail base.

What we don't claim

- **Short sample.** DVOL only goes back to March 2021.
- **Aggregated indices.** No full option surface, so no skew and no term structure.
- **Daily frequency.** Intraday data would refine realized volatility.
- **Asymmetric regime definition.** VIX for equity stress vs. realized volatility for crypto stress.

Take-aways

1. Implied volatility is a **risk-adjusted expectation**, not a raw forecast, in both markets.
2. Equity markets are **close to efficient in normal periods**; crypto markets are **not**.
3. The variance risk premium **widens in equity stress** and **compresses in crypto stress**. This asymmetry is only visible across markets.

Code, data and full pipeline:
`github.com/X9ClementX9/iv-rv-thesis`

Thank you.

Questions?

Replication materials

github.com/X9ClementX9/iv-rv-thesis